



数据库系统课程实验报告

实验名称:	数据更新
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1.实验目的

- 掌握数据库的特点（字体 熟练掌握单条记录和小批量数据插入的方法（INSERT）
- 熟练掌握使用子查询实现数据插入的方法(INSERT INTO… SUBQUERY)
- 熟练掌握数据修改和删除的方法(UPDATE,DELETE,TRUNCATE)
- 理解约束对数据更新的影响：

2.实验内容和步骤

(1) 为地区表 regions 新增一条记录：('5', 'Oceania')。

步骤如下：

```
sales=> INSERT INTO regions(region_id,region_name) VALUES(5,'Oceania');  
INSERT 0 1
```

(2) 将 countries 表中的国家名为 Australia 的 region_id 改为 5。

步骤如下：

```
sales=> UPDATE countries SET region_id=5 WHERE country_name='Australia';  
UPDATE 1
```

(3) 使用一条批量插入数据语句为 countries 表新增 5 条记录：

('NO','Norway','1'), ('ES','Spain','1'),('SE','Sweden','1'), ('PT','Portugal','1'),
('NZ','New Zealand','5')。

步骤如下：

```
sales=> INSERT INTO countries(country_id,country_name,region_id) VALUES('NO','Norway',1),('ES','Spain',1),('SE','Sweden',1),('PT','Portugal',1),('NZ','New Zealand',5);  
INSERT 0 5
```

(4) 创建一张名为 Asia_countries(country_id,country_name)的新表，其中字段为 countries 表中的同名字段。

步骤如下：

```

sales=> CREATE TABLE Asia_countries(
sales(> country_id CHAR(2) NOT NULL,
sales(> country_name VARCHAR(40) NOT NULL,
sales(> CONSTRAINT Asia_countries_PK PRIMARY KEY(country_id)
sales(> );
NOTICE: CREATE TABLE / PRIMARY KEY will create implicit index "asia_countries_pk" for table "asia_countries"
CREATE TABLE

```

(5) 将 countries 表中所有亚洲国家的数据插入到该表中。（要求使用插入子查询结果的方法实现）

步骤如下：

```

sales=> INSERT INTO Asia_countries(country_id,country_name) SELECT country_id,country_name FROM countries WHERE region_id=(SELECT region_id FROM regions WHERE region_name='Asia');
INSERT 0 5

```

(6) 创建一张名为 order_total(order_id,total_price)的视图，该视图存放每个订单号及其总价，其中 total_price 为总价，其值为数量 quantity 与单价 unit_price 乘积之和，order_id, quantity 和 unit_price 为 order_items 表中的同名字段。

步骤如下：

```

sales=> CREATE VIEW order_total AS SELECT order_id,SUM(quantity*unit_price) AS total_price FROM order_items GROUP BY order_id;
CREATE VIEW

```

(7) 查询 order_total 视图中订单号 order_id 为 97 的总价并记录该结果。

步骤如下：

```

sales=> SELECT total_price AS original_price FROM order_total WHERE order_id=97;
original_price
-----
61676319.0000
(1 row)

```

(8) 将 order_items 表中 product_id 为 99 的单价 unit_price 增加 4 元。

步骤如下：

```

sales=> UPDATE order_items SET unit_price=unit_price+4 WHERE product_id=99;
UPDATE 2

```

(9) 查询视图 `order_total` 中订单号 `order_id` 为 97 的总价，将其与第 (7) 步的结果进行比较，观察其异同。

步骤如下：

查询出的值较 (7) 有变化，说明第 (8) 步修改单价的操作影响到了总价的计算，视图中的值也会随之改变。

```
sales=> SELECT total_price AS alter_price FROM order_total WHERE order_id=97;
        alter_price
-----
61676511.0000
(1 row)
```

(10) 使用 `delete` 命令删除 `Asia_countries` 表中 `country_id` 为 IN 的记录。

步骤如下：

```
sales=> DELETE FROM Asia_countries WHERE country_id='IN';
DELETE 1
```

(11) 使用 `truncate` 命令清空 `Asia_countries` 表的所有记录。

步骤如下：

```
sales=> TRUNCATE TABLE Asia_countries;
TRUNCATE TABLE
```

(12) 删除 `Asia_countries` 表和视图 `order_total`。

步骤如下：

1.删除基本表

```
sales=> DROP TABLE Asia_countries;
DROP TABLE
```

2.删除视图

```
sales=> DROP VIEW order_total;
DROP VIEW
```

(13) 使用命令 \d employees 查看 employees 表的外码约束语句，包括 on delete cascade 选项。

步骤如下：

```
sales=> \d employees;
Table "sales.employees"
  Column      |      Type      | Modifiers
-----+-----+-----
 employee_id | integer        | not null
 first_name  | character varying(255) | not null
 last_name   | character varying(255) | not null
 email       | character varying(255) | not null
 phone       | character varying(50)  | not null
 hire_date   | timestamp(0) without time zone | not null
 manager_id  | integer        |
 job_title   | character varying(255) | not null
Indexes:
    "employees_pk" PRIMARY KEY, btree (employee_id) TABLESPACE pg_default
Foreign-key constraints:
    "fk_employees_manager" FOREIGN KEY (manager_id) REFERENCES employees(employee_id)
Referenced by:
    TABLE "employees" CONSTRAINT "fk_employees_manager" FOREIGN KEY (manager_id) REFERENCES employees(employee_id)
    TABLE "orders" CONSTRAINT "fk_orders_employees" FOREIGN KEY (salesman_id) REFERENCES employees(employee_id)
```

\d+ 能查询到更详细的信息，约束都没有 on delete cascade

```
sales=> \d+ employees;
Table "sales.employees"
  Column      |      Type      | Modifiers | Storage | Stats target | Description
-----+-----+-----+-----+-----+-----
 employee_id | integer        | not null | plain   |              |
 first_name  | character varying(255) | not null | extended |              |
 last_name   | character varying(255) | not null | extended |              |
 email       | character varying(255) | not null | extended |              |
 phone       | character varying(50)  | not null | extended |              |
 hire_date   | timestamp(0) without time zone | not null | plain   |              |
 manager_id  | integer        |          | plain   |              |
 job_title   | character varying(255) | not null | extended |              |
Indexes:
    "employees_pk" PRIMARY KEY, btree (employee_id) TABLESPACE pg_default
Foreign-key constraints:
    "fk_employees_manager" FOREIGN KEY (manager_id) REFERENCES employees(employee_id)
Referenced by:
    TABLE "employees" CONSTRAINT "fk_employees_manager" FOREIGN KEY (manager_id) REFERENCES employees(employee_id)
    TABLE "orders" CONSTRAINT "fk_orders_employees" FOREIGN KEY (salesman_id) REFERENCES employees(employee_id)
Has OIDs: no
Options: orientation=row, compression=no
```

(14) 查询 employees 表中 manager_id 为 1 的记录。

步骤如下：

```

sales-> SELECT * FROM employees WHERE manager_id=1;

```

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
2	Jude	Rivera	juderivera@examplecom	5151234568	2021-09-16 00:00:00	1	Administration Vice President
3	Blake	Cooper	blakecooper@examplecom	5151234569	2013-09-16 00:00:00	1	Administration Vice President
15	Rory	Kelly	rorykelly@examplecom	5151274561	2007-12-16 00:00:00	1	Purchasing Manager
21	Jaxon	Ross	jaxonross@examplecom	6501231234	2018-07-16 00:00:00	1	Stock Manager
22	Liam	Henderson	liamhenderson@examplecom	6501232234	2010-02-16 00:00:00	1	Stock Manager
23	Jackson	Coleman	jacksoncoleman@examplecom	6501233234	2001-05-16 00:00:00	1	Stock Manager
24	Callum	Jenkins	callumjenkins@examplecom	6501234234	2010-10-16 00:00:00	1	Stock Manager
25	Ronnie	Perry	ronnieperry@examplecom	6501235234	2016-11-16 00:00:00	1	Stock Manager
46	Ava	Sullivan	avasullivan@examplecom	1.14413E+13	2001-10-16 00:00:00	1	Sales Manager
47	Ella	Wallace	ellawallace@examplecom	1.14413E+13	2005-09-16 00:00:00	1	Sales Manager
48	Jessica	Woods	jessicawoods@examplecom	1.14413E+13	2010-03-16 00:00:00	1	Sales Manager
49	Isabella	Cole	isabellacole@examplecom	1.14413E+13	2015-10-16 00:00:00	1	Sales Manager
50	Mia	West	miawest@examplecom	1.14413E+13	2029-09-16 00:00:00	1	Sales Manager
102	Emma	Perkins	emmaperkins@examplecom	5151235555	2017-02-16 00:00:00	1	Marketing Manager

(14 rows)

(15) 修改 employees 表的外码约束，去掉外码约束中的 on delete cascade 选项，但保留原有的外码引用，即 manager_id 引用本表上的 employee_id。（可通过先删后建实现）

步骤如下：

1.删除原有约束

```

sales-> ALTER TABLE employees DROP CONSTRAINT fk_employees_manager;
ALTER TABLE

```

2.添加新约束

```

sales-> ALTER TABLE employees ADD CONSTRAINT fk_employees_manager FOREIGN KEY (manager_id) REFERENCES employees(employee_id);
ALTER TABLE

```

(16) 删除 employees 表中 employee_id 为 1 的记录，观察操作结果。

步骤如下：

1.删除

```

sales-> DELETE FROM employees WHERE employee_id=1;
ERROR: update or delete on table "employees" violates foreign key constraint "fk_employees_manager" on table "employees"
DETAIL: Key (employee_id)=(1) is still referenced from table "employees".

```

2.查看结果


```

sales=> SELECT * FROM employees WHERE manager_id=1;

```

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
2	Jude	Rivera	juderivera@examplecom	5151234568	2021-09-16 00:00:00	1	Administration Vice President
3	Blake	Cooper	blakecooper@examplecom	5151234569	2013-09-16 00:00:00	1	Administration Vice President
15	Rory	Kelly	rorykelly@examplecom	5151274561	2007-12-16 00:00:00	1	Purchasing Manager
21	Jaxon	Ross	jaxonross@examplecom	6501231234	2018-07-16 00:00:00	1	Stock Manager
22	Liam	Henderson	liamhenderson@examplecom	6501232234	2010-02-16 00:00:00	1	Stock Manager
23	Jackson	Coleman	jacksoncoleman@examplecom	6501233234	2001-05-16 00:00:00	1	Stock Manager
24	Callum	Jenkins	callumjenkins@examplecom	6501234234	2010-10-16 00:00:00	1	Stock Manager
25	Ronnie	Perry	ronnieperry@examplecom	6501235234	2016-11-16 00:00:00	1	Stock Manager
46	Ava	Sullivan	avasullivan@examplecom	1.14413E+13	2001-10-16 00:00:00	1	Sales Manager
47	Ella	Wallace	ellawallace@examplecom	1.14413E+13	2005-09-16 00:00:00	1	Sales Manager
48	Jessica	Woods	jessicawoods@examplecom	1.14413E+13	2010-03-16 00:00:00	1	Sales Manager
49	Isabella	Cole	isabellacole@examplecom	1.14413E+13	2015-10-16 00:00:00	1	Sales Manager
50	Mia	West	miawest@examplecom	1.14413E+13	2029-09-16 00:00:00	1	Sales Manager
102	Emma	Perkins	emmaperkins@examplecom	5151235555	2017-02-16 00:00:00	1	Marketing Manager

(14 rows)

(17) 修改 employees 表的外码约束，增加 on delete cascade 选项，即回到最初的外码约束状态。

步骤如下：

1.删除原有约束

```

sales=> ALTER TABLE employees DROP CONSTRAINT fk_employees_manager;
ALTER TABLE

```

2.添加新的约束

```

sales=> ALTER TABLE employees ADD CONSTRAINT fk_employees_manager FOREIGN KEY (manager_id) REFERENCES employees(employee_id) ON DELETE CASCADE;
ALTER TABLE

```

(18) 再次执行第 (16) 步，观察操作结果。

步骤如下：

1.再次执行结果依然没变，因为其他表的外键也是非 on delete cascade 约束。

```

sales=> DELETE FROM employees WHERE employee_id=1;
ERROR: update or delete on table "employees" violates foreign key constraint "fk_orders_employees" on table "orders"
DETAIL:  Key (employee_id)=(54) is still referenced from table "orders".

```

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
2	Jude	Rivera	juderivera@examplecom	5151234568	2021-09-16 00:00:00	1	Administration Vice President
3	Blake	Cooper	blakecooper@examplecom	5151234569	2013-09-16 00:00:00	1	Administration Vice President
15	Rory	Kelly	rorykelly@examplecom	5151274561	2007-12-16 00:00:00	1	Purchasing Manager
21	Jaxon	Ross	jaxonross@examplecom	6501231234	2018-07-16 00:00:00	1	Stock Manager
22	Liam	Henderson	liamhenderson@examplecom	6501232234	2010-02-16 00:00:00	1	Stock Manager
23	Jackson	Coleman	jacksoncoleman@examplecom	6501233234	2001-05-16 00:00:00	1	Stock Manager
24	Callum	Jenkins	callumjenkins@examplecom	6501234234	2010-10-16 00:00:00	1	Stock Manager
25	Ronnie	Perry	ronnieperry@examplecom	6501235234	2016-11-16 00:00:00	1	Stock Manager
46	Ava	Sullivan	avasullivan@examplecom	1.14413E+13	2001-10-16 00:00:00	1	Sales Manager
47	Ella	Wallace	ellawallace@examplecom	1.14413E+13	2005-09-16 00:00:00	1	Sales Manager
48	Jessica	Woods	jessicawoods@examplecom	1.14413E+13	2010-03-16 00:00:00	1	Sales Manager
49	Isabella	Cole	isabellacole@examplecom	1.14413E+13	2015-10-16 00:00:00	1	Sales Manager
50	Mia	West	miawest@examplecom	1.14413E+13	2029-09-16 00:00:00	1	Sales Manager
102	Emma	Perkins	emmaperkins@examplecom	5151235555	2017-02-16 00:00:00	1	Marketing Manager

(14 rows)

2.补充：理想情况下，初始两个外键和相关表外键应均为 on delete

cascade 约束，但可能前几次实验的误差，导致了问题，现将两个约束均改为 on delete cascade，再重复（16）

```
sales=> ALTER TABLE orders DROP CONSTRAINT fk_orders_employees;
ALTER TABLE

sales=> ALTER TABLE orders ADD CONSTRAINT fk_orders_employees FOREIGN KEY (salesman_id) REFERENCES employees(employee_id) ON DELETE CASCADE;
ALTER TABLE

sales=> ALTER TABLE order_items DROP CONSTRAINT fk_order_items_orders;
ALTER TABLE
sales=> ALTER TABLE order_items ADD CONSTRAINT fk_order_items_orders FOREIGN KEY(order_id) REFERENCES orders(order_id) ON DELETE CASCADE;
ALTER TABLE

sales=> DELETE FROM employees WHERE employee_id=1;
DELETE 1

sales=> SELECT * FROM employees WHERE manager_id=1;
 employee_id | first_name | last_name | email | phone | hire_date | manager_id | job_title
-----+-----+-----+-----+-----+-----+-----+-----
(0 rows)
```

同时，重复（13），可以观察到有 cascade

```
sales=> \d employees;
          Table "sales.employees"
   Column   |          Type          | Modifiers
-----+-----+-----+-----+-----+-----+-----+-----
 employee_id | integer                | not null
 first_name  | character varying(255) | not null
 last_name   | character varying(255) | not null
 email       | character varying(255) | not null
 phone       | character varying(50)  | not null
 hire_date   | timestamp(0) without time zone | not null
 manager_id  | integer                |
 job_title   | character varying(255) | not null
Indexes:
    "employees_pk" PRIMARY KEY, btree (employee_id) TABLESPACE pg_default
Foreign-key constraints:
    "fk_employees_manager" FOREIGN KEY (manager_id) REFERENCES employees(employee_id) ON DELETE CASCADE
Referenced by:
    TABLE "employees" CONSTRAINT "fk_employees_manager" FOREIGN KEY (manager_id) REFERENCES employees(employee_id) ON DELETE CASCADE
    TABLE "orders" CONSTRAINT "fk_orders_employees" FOREIGN KEY (salesman_id) REFERENCES employees(employee_id) ON DELETE CASCADE
```

3. 实验总结

3.1 完成的工作

根据实验内容与步骤依次完成相关数据插入、修改与删除，并补充由于此前实验造成的实验误差。

3.2 对实验的认识

通过本实验，熟悉了数据更新的基本操作，掌握了 update, alter, delete, drop 等的相关语法，并加深了对约束的认识。

关于实验思考题：当更新数据失败时，一个主要原因可能是因为违反了完整性约束，如主外码约束，唯一性约束等。问题：请设计实例来验证外码约束中的 on update cascade 选项的作用。

创建三个表，父表和两个子表，其中一个带 on update cascade，另一个不带。

```
sales=> CREATE TABLE Test_Parent(  
sales(> id INT PRIMARY KEY,  
sales(> name VARCHAR(10) NOT NULL  
sales(> );  
NOTICE: CREATE TABLE / PRIMARY KEY will create implicit index "test_parent_pkey" for table "test_parent"  
CREATE TABLE
```

```
sales=> CREATE TABLE test_child1(  
sales(> id INT PRIMARY KEY,  
sales(> name VARCHAR(100) NOT NULL,  
sales(> parent_id INT,  
sales(> CONSTRAINT fk1 FOREIGN KEY (parent_id) REFERENCES test_parent(id) ON UPDATE CASCADE  
sales(> );  
NOTICE: CREATE TABLE / PRIMARY KEY will create implicit index "test_child1_pkey" for table "test_child1"  
CREATE TABLE
```

```
sales=> CREATE TABLE test_child2(  
sales(> id INT PRIMARY KEY,  
sales(> name VARCHAR(100) NOT NULL,  
sales(> parent_id INT,  
sales(> CONSTRAINT fk2 FOREIGN KEY (parent_id) REFERENCES test_parent(id)  
sales(> );  
NOTICE: CREATE TABLE / PRIMARY KEY will create implicit index "test_child2_pkey" for table "test_child2"  
CREATE TABLE
```

插入数据。

```
sales=> INSERT INTO Test_Parent VALUES(1,'test1'),(2,'test2');  
INSERT 0 2  
sales=> INSERT INTO Test_Child1 VALUES(101,'t1',1),(102,'t2',1),(103,'t3',2);  
INSERT 0 3  
sales=> INSERT INTO Test_Child2 VALUES(101,'t1',1),(102,'t2',1),(103,'t3',2);  
INSERT 0 3
```

直接 update，遇到报错。

```
sales=> UPDATE Test_Parent SET id=10 WHERE id=1;  
ERROR: update or delete on table "test_parent" violates foreign key constraint "fk2" on table "test_child2"  
DETAIL: Key (id)=(1) is still referenced from table "test_child2".
```

数据也没有更新。

```
sales=> SELECT * FROM Test_Parent;
id | name
----+-----
 1 | test1
 2 | test2
(2 rows)
```

```
sales=> SELECT * FROM Test_Child1;
id  | name | parent_id
-----+-----+-----
101 | t1   |          1
102 | t2   |          1
103 | t3   |          2
(3 rows)
```

```
sales=> SELECT * FROM Test_Child2;
id  | name | parent_id
-----+-----+-----
101 | t1   |          1
102 | t2   |          1
103 | t3   |          2
(3 rows)
```

删除不带 on update cascade 的表。

```
sales=> DROP TABLE test_child2;
DROP TABLE
sales=> UPDATE Test_Parent SET id=10 WHERE id=1;
UPDATE 1
```

数据更新。

```

sales=> SELECT * FROM Test_Parent;
  id | name
----+-----
   2 | test2
  10 | test1
(2 rows)

sales=> SELECT * FROM Test_Child1;
  id | name | parent_id
----+-----+-----
 103 | t3   |         2
 101 | t1   |        10
 102 | t2   |        10
(3 rows)

```

验证得出通过 on update cascade，可以使得子表应用父表进行的更新。

3.3 遇到的困难及解决方法

由于前期操作的不成熟，在实验 1 和实验 2 创建外键约束时都没有添加 on delete cascade，通过补充说明补全了实验。